STATE BOARD SCIENCE 2: CURRICULUM ARCHITECTURE

Maharashtra State Board Class 10 SSC • Academic Regulations & Weightage Matrix

1. Preamble & Pedagogical Objectives

Science and Technology Part 2 encompasses Biological Sciences, Human Physiology, Genetics, Environmental Conservation, Green Energy Technologies, Social Health, and Disaster Management.

The curriculum aims to foster empirical scientific inquiry, ecological stewardship, biological literacy, and responsible societal citizenship in compliance with national educational standards.

2. Examination Pattern & Unit-Wise Marks Distribution

- Written Board Examination: 40 Marks (Time: 2 Hours)
- Internal Continuous Comprehensive Evaluation & Practicals: 10 Marks
- Total Science 2 Assessment: 50 Marks (Combined with Science 1 for a 100-mark composite science score)
- Unit 1: Genetics & Evolution (Chapters 1 & 2): 12 Marks with options
- Unit 2: Life Processes & Reproduction (Chapters 2 & 3): 14 Marks with options
- Unit 3: Ecology, Environment & Green Energy (Chapters 4 & 5): 12 Marks with options
- Unit 4: Animal Classification & Applied Microbiology (Chapters 6 & 7): 12 Marks with options
- Unit 5: Biotechnology, Social Health & Disasters (Chapters 8, 9 & 10): 12 Marks with options

CHAPTER 1: HEREDITY AND EVOLUTION

Core Unit 1 | Theory Hours: 14 | Evaluated Marks: 6 (without option), 8 (with option)

Syllabus Sub-Topics & Curricular Coverage

- 1.1 Hereditary mechanisms and genetic transmission from parental generation to offspring.
- 1.2 Central Dogma: Transcription, Translation, and Translocation mechanisms.
- 1.3 Mutation: Sudden sequence changes, Hugo de Vries theory, and sickle-cell gene alteration.
- 1.4 Evolution: Definition as gradual, progressive change across geological epochs.
- 1.5 Evidences of Evolution: Morphological, Anatomical (homologous bones), Vestigial organs (appendix, coccyx, ear muscles), Palaeontological & Carbon-14 dating, Connecting links (Peripatus, Platypus), and Embryological stages.
- 1.6 Theories of Evolution: Darwin’s Natural Selection, Survival of the Fittest, Origin of Species, Lamarckism (use/disuse & inheritance of acquired characters), and modern Speciation drivers.
- 1.7 Human Evolution: 7 crore years timeline from Lemur-like ancestors, Ramapithecus, Australopithecus, Neanderthal man to intelligent Homo sapiens.

Key Student Learning Outcomes (SLOs)

Students will be able to justify evolutionary homologies, trace human fossil progression, decipher codon-amino acid translation, and critique Lamarckian versus Darwinian evolutionary models.

CHAPTER 2 & 3: LIFE PROCESSES IN LIVING ORGANISMS

Energy Release, Cell Division & Reproduction Systems | Evaluated Marks: 12 (with option)

Chapter 2: Respiration, Nutrient Metabolism & Cell Division

- • Cellular respiration: Aerobic (Glycolysis / EMP pathway, TCA cycle / Krebs cycle, Electron Transfer Chain) vs Anaerobic fermentation.
- • Production of ATP currency through cellular oxidation of carbohydrates, proteins, and lipids.
- • Mitosis (Karyokinesis: Prophase, Metaphase, Anaphase, Telophase, and Cytokinesis) for somatic growth and repair.
- • Meiosis (Meiosis I and Meiosis II) with crossing over for genetic diversity in germ cells.

Chapter 3: Reproduction & Human Reproductive Health

- • Asexual reproduction: Binary fission, Multiple fission, Budding, Fragmentation, Regeneration, Vegetative propagation, and Spore formation.
- • Sexual reproduction in flowering plants: Pollination, Double fertilization, seed germination.
- • Human Male Reproductive System: Testes, scrotum, vas deferens, seminal vesicles, prostate, urethra.
- • Human Female Reproductive System: Ovaries, fallopian tubes, uterus, cervix, vagina, and Menstrual cycle hormonal cascades (FSH, LH, Estrogen, Progesterone).
- • Assisted Reproductive Technologies (ART): In-Vitro Fertilization (IVF), Surrogacy, and Sperm banks.

CHAPTER 4: ENVIRONMENTAL MANAGEMENT

Ecosystem Balance, Conservation Ecology & Solid Waste | Evaluated Marks: 7 (with option)

Core Conceptual Learning Units

- • Ecosystem components: Biotic (producers, consumers, decomposers) and Abiotic (soil, temperature, moisture, minerals).
- • Environmental pollution: Air, Water, and Soil contaminants and their biological magnification through food chains.
- • Biodiversity hot spots: Western Ghats, Indo-Burma, and Sundaland reserves.
- • Classification of Threatened Species: Endangered, Rare, Vulnerable, and Indeterminate species (IUCN Red List guidelines).
- • Sacred Groves (Devrai): Indigenous community-managed forest patches preserving endemic medicinal flora and sacred fauna.
- • Solid Waste Management: Principles of 3R (Reduce, Reuse, Recycle), composting, vermicomposting, and landfill hazards.

Environmental Ethics & Citizen Responsibilities

Focuses on individual ecological footprint reduction, tree plantation mandates, single-use plastic eradication, and citizen compliance with environmental protection acts.

CHAPTER 5: TOWARDS GREEN ENERGY

Sustainable Energy Generation, Renewable Technologies | Evaluated Marks: 6 (with option)

Comparative Energy Generation Systems

- • Principle of Electromagnetic Induction (Michael Faraday) utilized in rotating turbine generators.
- • Thermal Power Plants: Chemical energy in coal -> Thermal energy -> Kinetic energy in steam -> Mechanical energy in turbine -> Electrical energy. Environmental drawbacks: Greenhouse gases (CO_2), acid rain precursors (SOx, NOx), toxic ash disposal.
- • Nuclear Power Plants: Controlled fission of Uranium-235 or Plutonium-239. Merits: High energy yield without air pollution; Risks: Radioactive waste storage and catastrophic meltdown risks.
- • Hydroelectric Power Plants: Potential energy of dam water -> Kinetic energy -> Turbine -> Generator. Merits: Zero fuel combustion; Demerits: Submergence of fertile lands and rehabilitation issues.
- • Solar Energy Technologies: Photovoltaic solar cells (Silicon semiconductors) converting radiant photon energy directly into DC electricity; Solar thermal power plants concentrating sunlight to generate steam.

CHAPTER 6 & 7: ANIMAL CLASSIFICATION & APPLIED MICROBIOLOGY

Taxonomy Criteria & Industrial Biotechnology | Evaluated Marks: 10 (with option)

Chapter 6: Criteria for New Animal Classification

- • Levels of Organization: Cellular (Porifera), Tissue (Cnidaria), Organ, and Organ-System level (Arthropoda, Chordata).
- • Body Symmetry: Asymmetrical (Sponges), Radial (Hydra, Sea star), and Bilateral (Planaria, Fish, Humans).
- • Germ Layers: Diploblastic (Ectoderm, Endoderm) vs Triploblastic (Ectoderm, Mesoderm, Endoderm).
- • Body Cavity (Coelom): Acoelomate, Pseudocoelomate (Aschelminthes), and Eucoelomate.
- • Major Phyla: Porifera, Coelenterata, Platyhelminthes, Aschelminthes, Annelida, Arthropoda, Mollusca, Echinodermata, Hemichordata, and Chordata (Pisces, Amphibia, Reptilia, Aves, Mammalia).

Chapter 7: Applied Microbiology & Food Technology

- • Industrial production of dairy products: Yogurt, Cheese production using Lactobacillus bulgaricus and Streptococcus thermophilus.
- • Probiotics: Active bacterial supplements maintaining beneficial gut microbiome balance.
- • Microbial enzymes, amino acids, biofuel generation (ethanol from molasses), and bioremediation of oil spills using Alcanivorax borkumensis.

CHAPTER 8 & 9: CELL BIOLOGY, BIOTECHNOLOGY & SOCIAL HEALTH

Stem Cells, Gene Editing & Psychosocial Wellbeing | Evaluated Marks: 9 (with option)

Chapter 8: Cell Biology and Biotechnology

- • Stem Cells: Totipotent, Pluripotent, and Multipotent stem cells obtained from umbilical cord blood, blastocysts, and adult bone marrow for organ regeneration and regenerative medicine.
- • Genetic Engineering: Recombinant DNA technology, transgenic crops (Bt Cotton, Golden Rice enriched with Vitamin A), and therapeutic human insulin production.
- • Cloning: Reproductive cloning (Dolly the sheep) and therapeutic tissue cloning.

Chapter 9: Social Health & Digital Age Challenges

- • Factors affecting social health: Mental stress, addiction to tobacco, alcohol, gutkha, and narcotics.
- • Incurable diseases, social exclusion, and irrational superstitions.
- • Communication Technology & Cyber Hazards: Compulsive selfie clicking, mobile screen fatigue, cyberbullying, online gambling, and financial cybercrimes.
- • Stress Management Techniques: Laughter clubs, regular physical exercise, hobbies (music, reading, gardening), meditation, and professional psychological counseling (e.g., Childline 1098).

CHAPTER 10: DISASTER MANAGEMENT & EVALUATION BLUEPRINT

Emergency Protocols, Mock Drills & Board Blueprint Rubric | Evaluated Marks: 6 (with option)

Chapter 10: Disaster Classification & Response Framework

- • Disaster Categorization: Geological (Earthquakes, Volcanism, Landslides), Climatic (Cyclones, Floods, Cloudbursts, Drought), and Man-made (Industrial chemical gas leaks, Nuclear accidents, Riots, Forest fires).
- • Disaster Management Cycle: 1. Preparedness 2. Early Warning 3. Immediate Rescue 4. Relief Supplies 5. Rehabilitation & Reconstruction.
- • First Aid Protocols: ABC Principle (Airway, Breathing, Circulation), pressure bandages for arterial bleeding, RICE protocol for sprains, and burns neutralization.
- • Essential Disaster Signs & Universal Symbols: Exit routes, fire alarm pull stations, radiation hazard emblems, and inflammable liquid notices.

Science 2 Question Paper Weightage Blueprint Scheme

- • Objective Q.1 (A & B): 10 Questions × 1 Mark = 10 Marks (Compulsory)
- • Short Answer Q.2 (A & B): 5 Questions × 2 Marks = 10 Marks (Choice: 5 out of 7)
- • Conceptual Q.3: 5 Questions × 3 Marks = 15 Marks (Choice: 5 out of 7)
- • Long Answer Q.4: 1 Question × 5 Marks = 5 Marks (Choice: 1 out of 2)
- • Total Composite Marks: 40 Marks (60 Marks with full optional questions)